

Juan Pablo Silva Alvarado



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<https://ertwro.github.io/cv.pdf> 
(Last updated March 12, 2026.)

Basic Information

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🌐: ertwro.github.io

👤: github.com/ertwro
📺: youtube.com/c/ertwro
🐦: twitter.com/ertwro

Technical Experience & Independent Research

2024–PRESENT

Independent Research Scientist

- **Systems Architecture & Formal Methods:** Engineered a verified reduction of the $1/3$ – $2/3$ Poset Conjecture, isolating combinatorial failure points into specific topological structures using Lean 4. Demonstrated ability to achieve machine-verified (0-sorry) results by architecting formal reductions and utilizing AI-assisted formalization workflows.
- **Tool Development (Rust):** Built **KVAC** and **FEG_prism**, implementing custom logic for topological mass derivations and complex system modeling. Proved ability to build complex, custom tools from scratch in a rigorous systems language.
- **AI-Steered Formalization:** Developed a workflow to “turbocharge” formal verification by constraining LLMs to generate machine-checkable proofs, bypassing traditional manual formalization bottlenecks.

Professional Clinical Experience

2014–PRESENT

Clinical Physician

- **Human Systems & Diagnostic Routing:** Provided high-stakes medical diagnostics and surgical support in various Colombian health institutions (Compensar, Clínica Medical, Virrey Solis).
- **Decision-Tree Logic:** Applied rigorous diagnostic protocols to treat acute and sub-acute conditions, managing complex biological variables in real-time. Equivalent to high-stakes, real-world diagnostic routing.
- **Surgical Assistance:** Advanced surgical support in Orthopedics, Plastic Surgery, General Surgery, and Neurosurgery.

2021–2022

Coordinator of Hospital Physicians

Fundación Cardiovascular San Rafael Managed medical teams and hospital flow.

2020–2021

Hospital Physician (Electrophysiology)

Fundación Cardiovascular San Rafael Supervised outpatient and inpatient care in Electrophysiology.

Education

2007–2014	Doctor of Medicine (Médico Cirujano) <i>Universidad Nacional de Colombia, Bogotá</i> Emphasis on Internal Medicine rotations.
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Skills

MACHINE LANGUAGES	Systems: Rust (Proficient), Bash/Zsh, Python. Formal/Math: Lean 4 (Applied/Research-focused), LaTeX. Functional (Foundational): Haskell and Emacs Lisp.
HUMAN LANGUAGES	English (Fluent), Spanish (Native), Italian (A2), French (A2), Japanese & German (A1).

Selected Online Publications & Projects

FORMAL VERIFI- CATION & MATH RESEARCH	Verified reduction of the 1/3–2/3 Poset Conjecture in Lean 4. (Check repository at https://github.com/ertwro/one_third_two_thirds)
SCIENTIFIC OUTREACH	Authored numerous articles on Zero-Knowledge Proofs, Protein Design, and Quantum Mechanics at Steemit/Hive .

Interests

Formal Methods, AI Safety, Molecular Gastronomy, Algebraic Geometry, Brazilian Jiu-Jitsu, Rock Climbing, Piano.

References

Available upon request.